

PROJECTIONS OF SOLIDS

Karthik M S
Assistant Professor
Dept. of Mechanical Engineering
SJCE, JSS S&TU Mysore.

PROJECTIONS OF SOLIDS

PRISMS - EDGE RESTING - PROBLEMS

[1] A square prism 35mm sides of base and 65 mm axis length rests on HP on one of its edges

1st Position T V

of the base which is inclined to VP at 30° . Draw the projections of the prism when the axis

3rd Position T V

inclined to HP at 45° .

2nd Position F V

[2.a] A square prism 35mm sides of base and 65mm axis length rests on HP on one of its

1st Position T V

edges of the base. Draw the projection of the prism when the axis is inclined to HP at 45° and

2nd Position F V

3rd Position T V

VP at 30° . [Sketch book problem]

[2.b] Case 1: (Appears to be inclined to VP at 60°)

[3] A pentagonal prism of 25mm sides of base and 60mm axis length rests on HP on **one of its edges of the base** which is **inclined to VP at 30°** . Draw the projections of the prism when the

1st Position T V

3rd Position T V

axis is inclined to HP at 45° . [Sketch book problem]

2nd Position F V

[4.a] A pentagonal prism of 25mm sides of base and 60mm axis length rests on HP on **one of its edges of the base**. Draw the projection of the prism when the axis is inclined to **HP at 45°**

1st Position T V

2nd Position F V

and VP at 30° .

3rd Position T V

[4.b] Case 1: (Appears to be inclined to VP at 60°)

[5] A hexagonal prism 25mm sides of base and 50mm axis length rests on HP on one of its edges. Draw the projections of the prism when the axis is inclined to HP at 45° and appears to be inclined to VP 40° .

1st Position T V

2nd Position F V

3rd Position T V

[6] A hexagonal prism 25mm sides of base and 50mm axis length rests on HP on one of its edges of the base. Draw the projections of the prism when the axis is inclined to HP at 45° and VP at 30° .

1st Position T V

2nd Position F V

3rd Position T V

PRISMS - CORNER RESTING - PROBLEMS

[7] A square prism 35mm sides of base and 65mm axis length rests on HP on **one of its corners of the base** such that the two base edges containing the corner on which it rests make equal inclinations with HP. Draw the projections of the prism **when the axis of the prism is inclined to HP at 40° and to VP at 30°** .

1st Position T V

2nd Position F V

3rd Position T V

[8] A pentagonal prism of 25mm sides of base and 60mm axis length rests on HP on **one of its corners of the base** such that the two base edges containing the corner on which it rests makes equal inclinations with HP. Draw the projections of the prism when the axis of the prism is inclined to HP at 40° and VP at 30° . [**Sketch book problem**]

1st Position T V

2nd Position F V

3rd Position T V

[9] A hexagonal prism 25mm sides of base and 50mm axis length rests on HP on one of its corners of its base such that the two base edges containing the corner on which it rests makes equal inclinations with HP. Draw the projections of the prism when the axis of the prism is inclined to HP at 40° and VP at 30° .

1st Position T V

2nd Position F V

3rd Position T V

PYRAMIDS

[1] A square pyramid 35mm sides of base and 60mm axis length rests on HP on one of its

1st Position T V

slant edges. Draw the projections of the pyramid when the axis is inclined to VP at 45° .

2nd Position F V

3rd Position T V

Corner resting

[2] A hexagonal pyramid 25 mm sides of base and 50mm axis length rests on HP on one of its

1st Position T V

slant edges. Draw the projections of the pyramid when the axis appears to be inclined to VP

2nd Position F V

at 45° . [Sketch book problem]

3rd Position T V

Corner resting

[3] A square pyramid 35mm sides of base and 65mm axis length rests on HP on one of its slant triangular faces. Draw the projections of the pyramid when the axis is inclined to VP at 45° . [Sketch book problem]

1st Position T V
2nd Position F V
3rd Position T V

Edge resting

[4] A pentagonal pyramid of 25mm sides of base and 60mm axis length rests on HP on one of edges of the base which is inclined to VP at 30° . Draw the projections of the pyramid when it's the axis is inclined to HP at 40° .

1st Position T V
3rd Position T V
2nd Position F V

[5] A pentagonal pyramid of 25mm sides of base and 50mm axis length rests on HP on one of its slant triangular faces. Draw the projections of the pyramid when the axis is inclined to VP at 45° . [Sketch book problem]

1st Position T V
2nd Position F V
3rd Position T V

Edge resting

[6] A pentagonal pyramid of 25mm sides of base and 50mm axis length rests on HP on one of its slant edges. Draw the projections of the pyramid when the axis is inclined to VP at 45° .

2nd Position F V

1st Position T V

3rd Position T V

Corner resting

CONE

[1] A cone of base diameter 50mm and axis length 60mm is resting on HP on a point on the circumference of its base. Draw the projections when the axis is inclined to HP at 40° and VP at 30° . [Sketch book problem]

1st Position T V

2nd Position F V

3rd Position T V

[2] A cone of base diameter 50mm and axis length 60mm is resting on HP on one of its generators. Draw the projections when the axis is inclined to VP at 30° .

1st Position T V

2nd Position F V

3rd Position T V

CYLINDER

[1] A cylinder of base diameter 50mm and axis length 60mm is resting on HP on a point on

1st Position T V

the circumference of its base. Draw the projections when the axis is inclined to HP at 40° and

2nd Position F V

VP at 30° .

3rd Position T V

IMPORTANT CONCEPTS

TRUE ANGLE = TRUE LENGTH (edge or axis or diagonal or median or perpendicular bisector)
VP **2nd Position TV**

APPARENT ANGLE = APPARENT LENGTH (edge or axis or diagonal or median or perpendicular bisector)
VP **2nd Position TV**

TRUE ANGLE $\neq$ APPARENT LENGTH (edge or axis or diagonal or median or perpendicular bisector)
VP **2nd Position TV**

→ Calculate: **APPARENT ANGLE** β

APPARENT ANGLE $\neq$ TRUE LENGTH (edge or axis or diagonal or median or perpendicular bisector)
VP **2nd Position TV**

→ Calculate: **TRUE ANGLE** ϕ